



Multiple Choice:

- Two turtles A and B start at the same time move towards each other at a distance of 150 m. The rate of turtle A is 10 m/s while that B is 20 m/s. A fly flies from one turtle to another at the same time that the turtles start to move toward its each other. The rate of the fly is constant at 100 m/s. determine the total distance traveled by the fly until the two turtles met?
 - 500 m
 - 600 m
 - 700 m
 - 800 m
- If A's rate of doing work is to B's as 6:8 and A does a piece of work in 50 days, how long will it take B to do it?
 - 66.7 days
 - 23 days
 - 50.2 days
 - 37.5 days
- It is now between 3 and 4 o'clock and in 20 minutes, the minute hand will be as much as the hour hand as it is now behind it. What is the time now?
 - 3:06.36
 - 3:20
 - 3:26.46
 - 3:30
- Elizabeth is now 18 years old and her colleague Eumia is 14 years old. How many years ago was Elizabeth twice as old as Eumia?
 - 5
 - 7
 - 8
 - 10
- A father tells his son, "I was your age now when you were born." If the father is now 38 years old, how old was his son 2 years ago?
 - 15
 - 17
 - 19
 - 21
- A goldsmith has two alloys of gold, the first being 70% pure and the second being 60% pure. How many ounces of the 60% pure gold must be used to make 100 ounces of an alloy which will be 66% gold?
 - 40
 - 45
 - 35
 - 38
- A jogger starts a course at a steady rate of 8 kph. Five minutes later, a second jogger starts the same course at 10 kph. How long will it take the second jogger to catch the first?
 - 20 min
 - 21 min
 - 22 min
 - 18 min
- Six years ago, Raymond Reddington was five times as old as Elizabeth Keen. In five years, Raymond Reddington will be three times as old as Elizabeth Keen. What is the present age of Elizabeth Keen?
 - 17
 - 16
 - 15
 - 14
- A man on a boat rows to a place 4.8 miles with the stream and back in 14 hours, but finds that he can row 14 miles with the stream in the same time as 3 miles against the stream. Find the rate of the stream.
 - 0.45 miles per hour
 - 0.55 mile per hour
 - 0.76 mile per hour
 - 0.68 mile per hour
- 6-year-old Jan Nicole visited Santa at a local department store. He gave her this riddle: "I started working at 15. I spent 1/4 of my working life in a factory. I spent 1/5 of my working life in an office, and I spent 1/3 of my working life as a school caretaker. For the last 13 years of my working life I've been Santa Claus. How old am I?"
 - 67
 - 75
 - 71
 - 78
- Twice the middle digit of a three-digit number is the sum of the other two. If the number is divided by the sum of its digits, the answer is 56 and the remainder is 12. If the digits are reversed, the number becomes smaller by 594. Find the number.
 - 432
 - 147
 - 852
 - 258
- Ely walks down to the bottom of an escalator that is moving up and he counts 150 steps. Her friend, Mia walks up to the top of the escalator and counts 75 steps. If Ely's speed of walking (in steps per unit time) is three times Mia's walking speed, how many steps are visible on the escalator at a given time? Assume that this value is constant.
 - 120
 - 140
 - 130
 - 150
- If the man's age is a 3-digit number, his age is equal to product of two numbers. One of the numbers is the sum of the digits and the other number is product of the digits his age is?
 - 144
 - 123
 - 155
 - 112
- At what time between 4 and 5 o'clock will the hands of the clock be at right angle for the first time?
 - 4:6.42
 - 4:5.23
 - 4:5.45
 - 4:7.62
- Michael has some coins in his pocket consisting of dimes, nickels and pennies. He has two more nickels than dimes, and three times as many pennies as nickels. How many pennies does he have if the total value is 52 cents?
 - 20
 - 12
 - 15
 - 5
- Bob has a coin collection made up of pennies and nickels. If he has three times as many pennies as nickels and the total face value of the coins is



EXCEL FIRST REVIEW AND TRAINING CENTER, INC.

Cebu: JRT Bldg. Imus Ave, Cebu City | Davao: 2nd Fl. MERCO Bldg. Rizal St. cor. Bolton St.
Manila: CMFFI Bldg. R. Papa St. Sampaloc | Baguio: 4th Fl. De Guzman Bldg., Legarda Road

excel online[®]

Math 01-08 to Math 01-19
Algebra

- | | | |
|--|---|---|
| <p>\$416, how many nickels are in the collection?</p> <p>A. 5200 C. 5600
B. 5400 D. 5800</p> <p>17. Diophantus passed $\frac{1}{6}$ of his life in childhood, $\frac{1}{12}$ in youth, $\frac{1}{7}$ as a bachelor. Five years after his marriage was born a son who died 4 years before his father at half his father's final age. How old was Diophantus when he died?</p> <p>A. 48 yrs. old C. 72 yrs. old
B. 84 yrs. old D. 27 yrs. old</p> <p>18. A reservoir can be filled by an inlet pipe in 24 hours and emptied in 28 hours. The foreman starts to fill the reservoir, but he forgets to close the outlet pipe. Six hours late he remembers and closes the outlet. How long does it take altogether to fill the reservoir?</p> <p>A. 29 $\frac{1}{7}$ hours C. 30 $\frac{1}{2}$ hours
B. 29 $\frac{1}{5}$ hours D. 30 $\frac{3}{5}$ hours</p> <p>19. A cyclist traveled a certain distance at her usual speed. If her speed had been 2 mi/hr. faster, she could have travelled the distance in 1 hour less time. If her speed had been 2 mi/hr. slower, she would have taken 2 hours longer. Find her usual speed.</p> <p>A. 6 mph C. 8 mph
B. 7 mph D. 9 mph</p> <p>20. The ratio of men to women in a certain part was 5:3. Twenty-four men got drunk and decided to leave the party. When twenty-four more women joined the party, the ratio between men to women was interchanged, how many men were left in the party?</p> <p>A. 24 C. 36
B. 15 D. 42</p> <p>21. The electric power which a transmission line can transmit is proportional to the product of its design voltage and current capacity, and inversely to the transmission distance. A 115 kV line rate at 1000</p> | <p>A can transmit 150 MW over 150 km. How much power in MW can 230 kV line rated at 1500 A transmit over 100 km?</p> <p>A. 675 MW C. 576 MW
B. 765 MW D. 567 MW</p> <p>22. Once a month, a man puts some money into the cookie jar. Each month he puts 50 centavos more into the jar than the month before. After 12 years, he counted his money, he had P 5,436. How much money did he put in the jar in the last month?</p> <p>A. P 73.50 C. P 75.50
B. P 74.50 D. P 72.50</p> <p>23. Determine x so that: x, $2x + 7$, $10x - 7$ will be a geometric progression.</p> <p>A. 7, $-\frac{7}{12}$ C. 7, $-\frac{5}{6}$
B. 7, $-\frac{14}{5}$ D. 7, $-\frac{7}{6}$</p> <p>24. Find the sum of the infinite geometric progression 6, -2, $\frac{2}{3}$, ...</p> <p>A. $\frac{9}{2}$ C. $\frac{5}{2}$
B. $\frac{7}{2}$ D. $\frac{11}{2}$</p> <p>25. A rubber ball is made to fall from a height of 50 feet and is observed to rebound $\frac{2}{3}$ of the distance it falls. How far will the ball travel before coming to rest if the ball continues to fall in this manner?</p> <p>A. 200 feet C. 225 feet
B. 250 feet D. 275 feet</p> <p>26. The 3rd term of a harmonic progression is 15 and the 9th term is 6. Find the 11th term.</p> <p>A. 4 C. 5
B. 6 D. 7</p> <p>27. To conserve energy due to the present energy crisis, the Meralco tried to re-adjust their charges to electrical energy users who consume more than 2000 kw-hrs. For the first 100 kw-hr, they charged 40 centavos and increasing at a constant rate more than the preceding one until the fifth 100 kw-hr, the charge is 76 centavos.</p> | <p>How much is the average charge for the electrical energy per 100 kw-hr?</p> <p>A. 58 centavos C. 60 centavos
B. 62 centavos D. 64 centavos</p> <p>28. If one third of the air in a tank is removed by each stroke of an air pump, what fractional part of the total air is removed in 6 strokes?</p> <p>A. 0.7122 C. 0.9122
B. 0.6122 D. 0.8122</p> <p>29. Eight men can dig 150 ft of trench in 7 hours. Three men can backfill 100 ft of the trench in 4 hours. How long will it take 10 men to dig and backfill 200 ft of trench?</p> <p>A. 9.87 hours C. 9.12 hours
B. 9.45 hours D. 9.61 hours</p> <p>30. Atan can paint the room in 15 hours, Marlon can paint 50 percent faster than Atan, and Danny can paint twice as fast as Atan. Atan begins to paint the room and works alone for the first hour and a half. Then Marlon joins Atan, and they work together until half the room is painted. Then Danny joins Atan and Marlon, and they work together until the entire room is painted. Find the number of minutes after Atan begins for the three of them to finish painting the room.</p> <p>A. 334 C. 347
B. 321 D. 312</p> <p>31. A man rows downstream at the rate of 5 miles per hour and rows upstream at the rate of 2 miles per hour. How far downstream should he go if he is to return in $\frac{7}{4}$ hours after leaving?</p> <p>A. 2.5 miles C. 3.3 miles
B. 3.1 miles D. 2.7 miles</p> <p>32. The tens' digit of a number is 3 less than the units' digit. If the number is divided by the sum of the digits, the quotient is 4 and the remainder is 3. What is the original number?</p> <p>A. 36 C. 47
B. 58 D. 69</p> |
|--|---|---|



33. If 3 is subtracted from the numerator of a certain fraction, the value of the fraction becomes $\frac{3}{5}$. If 1 is subtracted from the denominator of the same fraction, it becomes $\frac{2}{3}$. Find the original fraction.

A. $\frac{35}{55}$ C. $\frac{36}{55}$
B. $\frac{3}{7}$ D. $\frac{32}{41}$

34. How much water must be evaporated from 10 kg solution which has 4% salt to make a solution of 10% salt?

A. 4 kg C. 5 kg
B. 6 kg D. 7 kg

35. A and B can do a piece of work in 42 days, B and C in 31 days and C and A in 20 days. In how many days can all of them do the work together?

A. 19 C. 17
B. 21 D. 15

36. It takes Jimmy twice as long as Sonny to do a certain piece of work. Working together, they can finish the work in 6 hours. How long would it take Sonny to do it alone?

A. 9 hours C. 18 hours
B. 12 hours D. 14 hours

37. If A can do the work in "x" days and B in "y" days, how long will they finish the job working together?

A. $\frac{x+y}{xy}$ C. $\frac{x+y}{2}$
B. $\frac{xy}{x+y}$ D. $\sqrt{xy}$

38. A pump can pump out water from a tank in 11 hours. Another pump can pump out water from the same tank in 20 hours. How long will it take both pumps to pump out the water in the tank?

A. 7 hours C. 6 hours
B. $7\frac{1}{2}$ hours D. $6\frac{1}{2}$ hours

39. A girl is one-third as old as her brother and 8 years younger than her

sister. The sum of their ages is 38 years. How old is the girl?

A. 4 C. 5
B. 6 D. 7

40. What is the sum of three arithmetic means between 34 and 42?

A. 114 C. 124
B. 134 D. 144

41. The electric power which a transmission line can transmit is proportional to the product of its design voltage and current capacity, and inversely to the transmission distance. A 115-kilovolt line rated at 100 amperes can transmit 150 megawatts over 150 km. How much power, in megawatts can a 230 kilovolt line rated at 150 amperes transmit over 100 km?

A. 785 C. 485
B. 675 D. 595

42. If x varies directly as y and inversely as z, and $x = 14$ when $y = 7$ and $z = 2$, find the value of x when $y = 16$ and $z = 4$.

A. 14 C. 4
B. 16 D. 8

43. What time after 3 o'clock will the hands of the clock be together for the first time?

A. 3:02.30 C. 3:17.37
B. 3:14.32 D. 3:16.36

44. A tank can be filled in 9 hours by one pipe, 12 hours by a second pipe and can be drained when full by a third pipe in 15 hours. How long will it take to fill an empty tank with all pipes in operation?

A. 7 hours and 12 minutes
B. 7 hours and 32 minutes
C. 7 hours and 42 minutes
D. 7 hours and 50 minutes

45. A man sold half an egg more than half his eggs. He then sold half an egg more than his remaining eggs. He did the same for the third time. If he

had 7 eggs left, how many eggs does he originally have?

A. 67 C. 65
B. 63 D. 61

46. Cathy left her house on a bicycle heading north at 8 mph. At the same time, her sister Nina headed south 12 mph. how long will it take for them to be 24 miles apart?

A. 1.5 hr C. 1.2 hr
B. 2.43 hr D. 1.3 hr

47. A collection of 44 coins containing pennies, dimes and quarters is worth \$5.90. There are twice as many dimes as quarters. How many pennies are in the collection?

A. 6 C. 8
B. 5 D. 4

48. A girl on a bicycle coasts downhill covering 4 feet the first second, 12 feet the second second, and in general, 8 feet more each second than the previous second. If she reaches the bottom at the end of 14 seconds, how far did she coasts?

A. 782 feet C. 780 feet
B. 784 feet D. 786 feet

49. An Excel reviewee got a score of 30% on Test 1 of the five number test in Algebra. On the last number he got 90% in which a constant difference more on each number that he had on the immediately preceding one. What was his average score in Algebra?

A. 50 C. 55
B. 60 D. 65

50. Determine the sum of the infinite

series: $S = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots + \left(\frac{1}{3}\right)^n$

A. $\frac{4}{5}$ C. $\frac{3}{4}$
B. $\frac{2}{3}$ D. $\frac{1}{2}$

End